

Quantum Gravitation

Problem Set 7

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Topic

Source Coupling and Scattering Amplitudes

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Source Coupling and Scattering Amplitudes. State all assumptions and conventions.

$$\mathcal{L}_{\text{int}} = -(\kappa/2)h_{\mu\nu}(T^{\mu\nu} + S^{\mu\nu})$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\mathcal{M} \sim \kappa^2 T_1 \Pi^{(2)} T_2 / (q^2 + m_g^2)$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$V(r) = -Gm_1 m_2 e^{-m_g c r / \hbar} / r$$

Show enough intermediate work that another student can reproduce the calculation.